



Maclaurin series

Consider a continuous function f which can be differentiated infinitely many times.
The **Maclaurin series expansion** of $f(x)$ is

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \dots = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)x^k}{k!}$$

The n th degree **Maclaurin polynomial** approximation to f is $M_n(x) = \sum_{k=0}^n \frac{f^{(k)}(0)x^k}{k!}$.



- a** Find the Maclaurin series representation for $f(x) = \frac{1}{2+x}$.
- b** Hence find $M_4(x)$, the 4th order Maclaurin polynomial approximation for $\frac{1}{2+x}$. Compare $M_4(1)$ with $f(1)$.

Step 1: Find the derivatives:

$$(a) f(x) = \frac{1}{2+x} = (2+x)^{-1}$$

$$\therefore f'(x) = (-1)(2+x)^{-2} = (-1)^1 1! (2+x)^{-2}$$

$$f''(x) = (-1)(-2)(2+x)^{-3} = (-1)^2 2! (2+x)^{-3}$$

$$f'''(x) = (-1)(-2)(-3)(2+x)^{-4} = (-1)^3 3! (2+x)^{-4}$$

$$\vdots$$

$$f^{(k)}(x) = (-1)^k k! (2+x)^{-(k+1)}$$



Step 2: Find each derivative at $x=0$

$$(a) f(x) = \frac{1}{2+x} = (2+x)^{-1} \longrightarrow f(0) = \frac{1}{2}$$

$$f'(x) = (-1)^1 1! (2+x)^{-2} \longrightarrow f'(0) = \frac{(-1)^1 1!}{2^2}$$

$$f''(x) = (-1)^2 2! (2+x)^{-3} \longrightarrow f''(0) = \frac{(-1)^2 2!}{2^3}$$

$$f'''(x) = (-1)^3 3! (2+x)^{-4} \longrightarrow f'''(0) = \frac{(-1)^3 3!}{2^4}$$

• • •

$$\therefore f^{(k)}(0) = \frac{(-1)^k k!}{2^{k+1}} \text{ for all } k \in \mathbb{Z}^+.$$



Step 3: Substitute into the formula:

Since $f(0) = \frac{1}{2}$, the Maclaurin series representation for $f(x)$ is

$$\begin{aligned} f(x) &= \frac{1}{2} + \sum_{k=1}^{\infty} \frac{(-1)^k \cancel{k!}}{\cancel{k!} 2^{k+1}} x^k \\ &= \frac{1}{2} + \sum_{k=1}^{\infty} \frac{(-1)^k}{2^{k+1}} x^k = \sum_{k=0}^{\infty} \frac{(-1)^k}{2^{k+1}} x^k \end{aligned}$$



b The 4th order Maclaurin approximation for $\frac{1}{2+x}$ is

$$\begin{aligned}M_4(x) &= \sum_{k=0}^4 \frac{(-1)^k}{2^{k+1}} x^k \\&= \frac{1}{2} + \frac{(-1)^1}{2^2} x + \frac{(-1)^2}{2^3} x^2 + \frac{(-1)^3}{2^4} x^3 + \frac{(-1)^4}{2^5} x^4 \\&= \frac{1}{2} - \frac{1}{4} x + \frac{1}{8} x^2 - \frac{1}{16} x^3 + \frac{1}{32} x^4\end{aligned}$$

$$\therefore M_4(1) = \frac{1}{2} - \frac{1}{4} + \frac{1}{8} - \frac{1}{16} + \frac{1}{32} = \frac{11}{32}$$

End of Lesson

Note: You can download this pdf from the link given in description.